Source IP & Destination IP

Every network packet has a **Source IP address** and a **Destination IP address**.

- The **Source IP** shows where the packet is coming from.
- The **Destination IP** shows where the packet is going.

For example, when I open a website:

- My computer's IP address is the **source IP**.
- The web server's IP address is the **destination IP**.

This addressing system allows data to travel from one device to another across a network.

Ports

Ports are like **doors on a computer**.
They tell the operating system **which service or application should receive the incoming data**.

For example:

- Web traffic uses **port 80** for HTTP.
- Secure web traffic uses **port 443** for HTTPS.

Ports help multiple services run on the same IP address without confusion.

Port Hierarchy (Well-known Ports)

Some ports are reserved for specific services and are called **well-known ports**:

- HTTP → 80
- HTTPS → 443
- FTP → 21
- SSH → 22
- DNS → 53
- SMTP → 25

These ports are standardized so that clients and servers know which service is running where.

Network Interfaces (Using ifconfig Command)

Go to the terminal and type:

ifconfig

You will see multiple network interfaces such as:

enp2s0 / eth0
This is the **Ethernet interface**.
If your PC is connected using an Ethernet cable, this interface will show an IP address.

lo
This is the **Loopback interface**.
It is used for **internal communication within the same system**.
Its IP address is usually 127.0.0.1.

wlp1s0 / wlan0
This is the **Wireless (Wi-Fi) interface**.
If connected to Wi-Fi, this interface will display an IP address.

proton0 / tun0 / wg0
These interfaces appear when you are connected to a **VPN**.

- tun0 → OpenVPN
- wg0 → WireGuard VPN

NOTE:
You may see different interface names depending on your operating system.
My examples are based on **Linux**.

Wireshark Interface Setup

Before capturing packets, it is good to configure Wireshark.

1. Open Wireshark.
2. Right-click on **Profile** and create a new profile if you want.
3. Go to **Edit → Preferences**.
4. You will see options such as Capture, Appearance, Coloring Rules, Protocols, etc.
5. Enable features according to your preference.

If you are using a newer version of Wireshark, you may see an option called **Packet Diagram**, which is very useful for learning.

After configuration, choose an interface and start capturing packets.

TCP Packet Capturing

TCP (Transmission Control Protocol) is a **connection-oriented protocol**.

It uses a **3-Way Handshake** to establish a connection:

1. SYN

2. SYN-ACK
3. ACK

Display Filters for TCP Handshake

- SYN Packet
  tcp.flags.syn == 1 && tcp.flags.ack == 0
- SYN-ACK Packet
  tcp.flags.syn == 1 && tcp.flags.ack == 1
- ACK Packet
  tcp.flags.syn == 0 && tcp.flags.ack == 1

These filters allow you to observe how TCP connections are established.

UDP (User Datagram Protocol)

Username: test
Password: 12345

Press Login.

While doing this, keep Wireshark capturing.

Filter HTTP Packets

http

or

http.request.method == "GET"

This shows GET requests.

Filter POST Requests

http.request.method == "POST"

Click the POST packet.

Expand:

- Frame
- Ethernet
- IP
- TCP
- HTTP

Scroll down and expand **HTML Form URL Encoded**.

You will see:

username=test
password=12345

This means the credentials are visible in plain text.

Conclusion on HTTP Security

HTTP does not encrypt data.
Anyone capturing traffic can read usernames and passwords.

This is why modern websites use **HTTPS**, which encrypts communication.

Video Demonstration

https://www.youtube.com/watch?si=Cvz90W1Lsgan-muV&v=aMu-hqTaspc&feature=youtu.be